

OSM / Opto-PING Rung 0N

K-Null Calibration Repair and Sensitivity-Specificity Balance v0.1

Status: PASS_NULL_CALIBRATION_REPAIRED

This is a synthetic calibration gate. It does not analyze human data and does not prove OSM or any biological mechanism.

Executive Verdict

Rung 0N repaired the sensitivity/specificity balance exposed by Rung 0M. The final decision now requires both held-out full-model prediction and calibrated K significance. Raw K alone is not enough; full-model prediction alone is not enough. The result is a synthetic calibration pass. It should be followed by a locked Rung 0O replication using fresh seeds without further tuning.

Locked Specification

Rung 0N spec SHA-256	f11cc9662d5a666c10f29e11f70c88a04e377e6499e87ae0bb1f1632e884a6b8
Selected full-model penalty	0.04
Total synthetic trials	410
Positive-control trials	60
Null / empty-sky trials	350

Core Decision Rule

FinalFullPlusKLock = FullPredictiveWin AND KSignificant
selected_threshold = max(generator_q99, pooled_q975)

K significance is interpreted only when the full reciprocal model also wins prediction. This is the main null-aware repair.

Main Numerical Results

Positive full+K lock rate	0.967
Positive K significant rate	1.000
Positive K within 15% rate	1.000
Positive K within 10% rate	0.950
Positive median K_hat_scaled	1.014
Null false full+K lock rate	0.000
Null raw K significant rate	0.011
Null interpreted K rate	0.000
Null full predictive win rate	0.000
Null non-full predictive win rate	1.000

Generator-Level Summary

Generator	n	Median K	Raw K sig	Full win	Interpreted K	Final lock	Dominant model
colored_noise_null	50	0.142	0.020	0.000	0.000	0.000	sensory_drive_only
generic_hidden_null	50	0.140	0.060	0.000	0.000	0.000	null_no_coupling

Generator	n	Median K	Raw K sig	Full win	Interpreted K	Final lock	Dominant model
permuted_label_null	50	0.055	0.000	0.000	0.000	0.000	sensory_drive_only
replay_only_null	50	0.056	0.000	0.000	0.000	0.000	null_no_coupling
sensory_drive_only_null	50	0.079	0.000	0.000	0.000	0.000	colored_noise_alt
standard_ping_null	50	0.059	0.000	0.000	0.000	0.000	sensory_drive_only
time_reversed_null	50	0.051	0.000	0.000	0.000	0.000	standard_ping
full_reciprocal_positive	60	1.014	1.000	0.967	0.967	0.967	full_reciprocal

K Null Calibration

Generator	Generator q99	Pooled q975	Selected threshold
colored_noise_null	0.342	0.214	0.342
generic_hidden_null	0.274	0.214	0.274
permuted_label_null	0.131	0.214	0.214
replay_only_null	0.106	0.214	0.214
sensory_drive_only_null	0.152	0.214	0.214
standard_ping_null	0.124	0.214	0.214
time_reversed_null	0.111	0.214	0.214

Penalty Tuning

Penalty	Positive final	Null false final	Raw null K sig	Score
0.000	1.000	0.000	0.011	1.000
0.020	1.000	0.000	0.011	1.000
0.040	0.967	0.000	0.011	0.967
0.060	0.900	0.000	0.011	0.900
0.080	0.700	0.000	0.011	0.625
0.100	0.467	0.000	0.011	0.275
0.120	0.300	0.000	0.011	0.025
0.140	0.150	0.000	0.011	-0.200
0.160	0.033	0.000	0.011	-0.375
0.180	0.017	0.000	0.011	-0.400
0.200	0.017	0.000	0.011	-0.400

Lock Decision Table

criterion	value	target	pass
positive_full_plus_K_lock_rate >= 0.80	0.967	0.800	True
positive_K_significant_rate >= 0.90	1.000	0.900	True
positive_K_within_15_rate >= 0.90	1.000	0.900	True
null_false_full_plus_K_lock_rate <= 0.05	0.000	0.050	True
null_interpreted_K_rate <= 0.05	0.000	0.050	True
null_non_full_predictive_win_rate >= 0.85	1.000	0.850	True

Interpretation

Rung 0N produced a cleaner sensitivity/specificity balance than Rung 0M. The positive-control generator locked in 96.7% of trials, while the empty-sky false final-lock rate was 0.0%. Some null trials still produced raw K-like values, but none became interpreted K events because the full reciprocal model did not win prediction.

This means the null-aware model selection rule is doing what it was designed to do: it refuses to let K act as a free-standing target detector.

Boundary

Synthetic sensitivity/specificity calibration only. No biological, human EEG, microtubular, biophotonic, or OSM mechanism claim.